# Geometry Visualizer

## PROJECT REPORT

**Submitted for CBCS in B.Tech – Computer Science and Engineering(CPS)**

**CSE203E WEB-PROGRAMMING**

**By**

**Pranav Bhatia – 24BPS1001**

**Slot: L23+L24+L37+L38**

**Faculty Name: Dr S Vigneshwari (54147)**

**School of Computer Science and Engineering**

**April-2026**

# Contents

**ABSTRACT** **2**

**1 INTRODUCTION** **3**

**2 TOOLS USED** **4**

**3 NOVELTY** **5**

**4 IMPLEMENTATION DETAILS** **6**

4.B OUTPUT SCREENSHOTS . . . . . 7

4.C DEMO LINK . . . . . 8

4.D GITHUB / WEBSITE HOSTING LINK . . . . . 8

**5 SOCIETAL OUTCOME OF THE PROJECT** **9**

**6 CONCLUSION AND FUTURE SCOPE** **10**

**REFERENCES** **11**

# ABSTRACT

The Geometry Visualizer is a modern web-based application developed using React and HTML5 Canvas that enables interactive visualization of geometric shapes and mathematical functions. Traditional learning methods often lack visual intuition, making mathematical concepts difficult to grasp. This project addresses that limitation by providing a real-time, dynamic environment where users can manipulate parameters and observe immediate graphical outputs.

The system supports a wide variety of geometric shapes such as circles, polygons, ellipses, and triangles, along with function plotting including trigonometric and exponential functions. A key feature of the application is the ability to input custom mathematical equations, which are parsed and rendered dynamically.

The application is designed with a modular architecture, ensuring scalability and maintainability. It also incorporates educational content such as formulas, properties, and mathematical facts to enhance conceptual understanding.

# Chapter 1

# INTRODUCTION

Mathematics, particularly geometry and graphing, plays a fundamental role in science and engineering. However, many learners struggle with understanding abstract concepts due to lack of visualization tools.

The Geometry Visualizer aims to bridge this gap by offering an intuitive and interactive interface where users can explore shapes and functions visually. Unlike static diagrams in textbooks, this application allows real-time parameter adjustments such as radius, angle, and frequency.

The system is built using modern frontend technologies, making it lightweight, fast, and accessible through any web browser. The integration of graphical rendering with user input enhances both learning and experimentation.

# Chapter 2

# TOOLS USED

**1. React 18:** Used for building the user interface with reusable components and efficient state management.

**2. Vite:** A fast build tool that enables quick development and hot module replacement.

**3. HTML5 Canvas API:** Used for rendering shapes and graphs with high performance and flexibility.

**4. JavaScript (ES6+):** Handles logic, mathematical computations, and event handling.

**5. CSS3:** Provides styling with a modern dark theme using custom variables.

# Chapter 3

# NOVELTY

The key innovations of this project include:

- Real-time visualization with instant updates
- Combination of geometry + function graphing in one platform
- Custom equation parsing and plotting
- Interactive parameter controls (sliders and inputs)
- Educational integration (formulas, properties, facts)
- Modern UI with dark theme and smooth UX

Unlike traditional tools, this system emphasizes both learning and interactivity.

# Chapter 4

# IMPLEMENTATION DETAILS

## 4.A CODE SNIPPETS

**Canvas Rendering Logic:**

```
// Drawing a circle using Canvas API
ctx.beginPath();
ctx.arc(centerX, centerY, radius, 0, 2 * Math.PI);
ctx.stroke();
```

**Circle Calculations:**

```
// Calculating properties of a circle
const area = Math.PI * radius * radius;
const circumference = 2 * Math.PI * radius;
const diameter = 2 * radius;
```

**Equation Handling:**

```
// Example function for plotting
const y = amplitude * Math.sin(frequency * x);
```

**React State Management:**

```
// Managing dynamic parameters
const [radius, setRadius] = useState(50);

<input
  type="range"
  value={radius}
  onChange={(e) => setRadius(e.target.value)}
/>
```

# 4.B OUTPUT SCREENSHOTS

The following screenshots demonstrate the working of the application interface and its features:

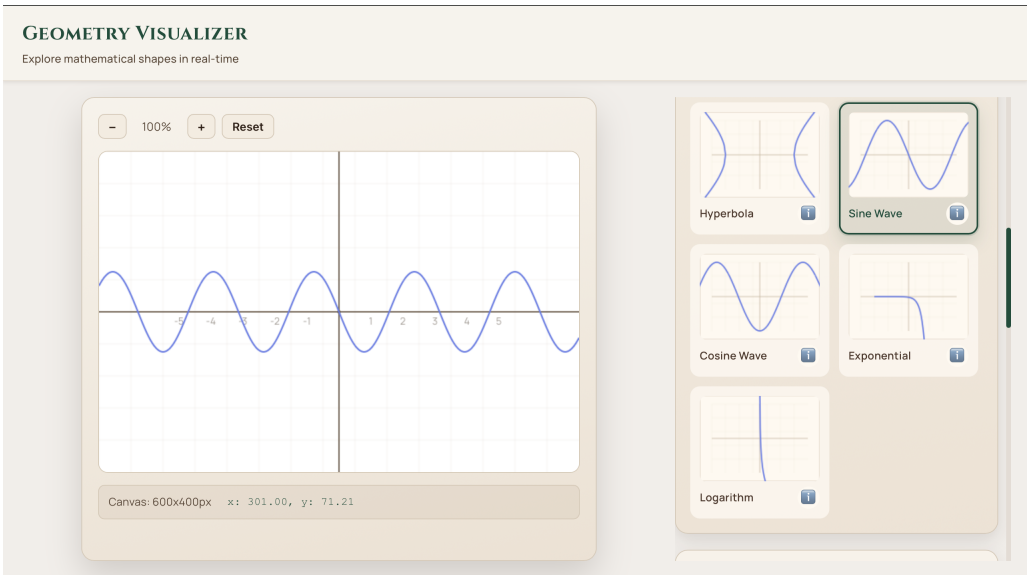


Figure 4.1: Function Selection Panel with Multiple Graph Options

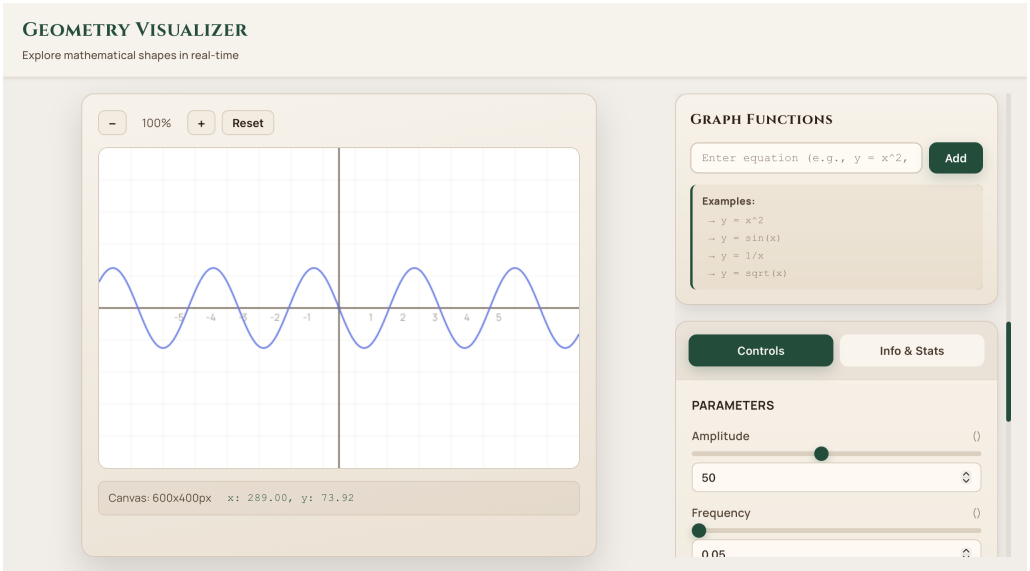


Figure 4.2: Sine Wave Graph Plotting with Parameter Controls

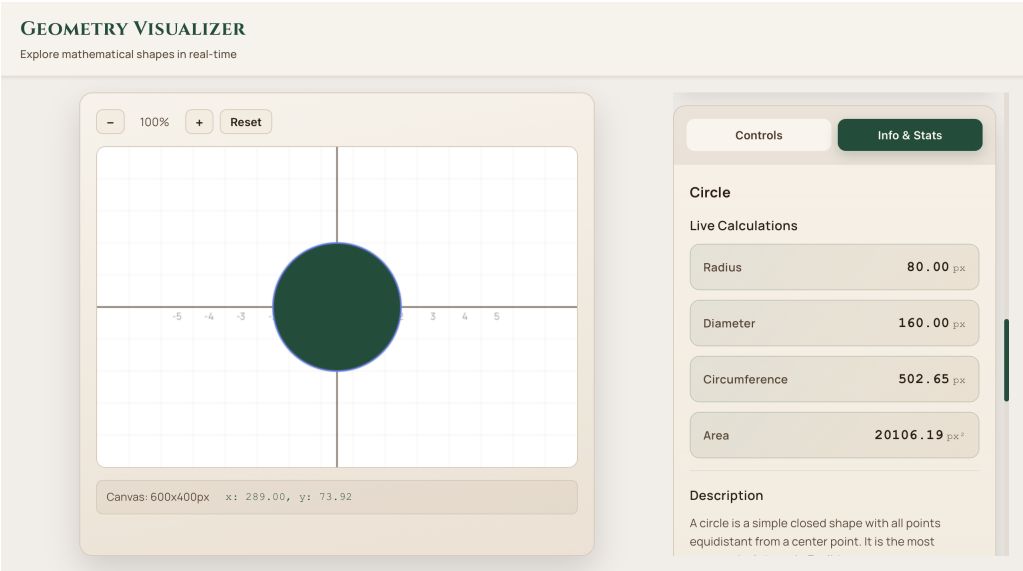


Figure 4.3: Interactive Circle Visualization with Real-Time Calculations

# 4.C DEMO LINK

The live demo of the application can be accessed at:

https://drive.google.com/file/d/1ubapj2JYV5-_bmlVbJDsHYmA-n35XftH/view?usp=sharing

# 4.D GITHUB LINK

The complete source code of the project is available at:

https://github.com/Pranav-player/Geometry-Visualizer-Web-

# Chapter 5

# SOCIETAL OUTCOME OF THE PROJECT

This project contributes to education by making mathematical concepts easier to understand through visualization. It can be used by students, teachers, and self-learners to improve conceptual clarity.

The platform encourages interactive learning, which is proven to be more effective than passive methods. It can also be extended for use in online education platforms and digital classrooms.

# Chapter 6

# CONCLUSION AND FUTURE SCOPE

The Geometry Visualizer successfully demonstrates how technology can enhance the learning experience. By combining interactivity with mathematical computation, the project makes abstract concepts more intuitive.

**Future Scope:**

- 3D visualization support
- Mobile application development
- AI-based equation suggestions
- Advanced graph analysis tools

# REFERENCES


- React Documentation: https://react.dev
- MDN Canvas API: https://developer.mozilla.org
- GitHub Repository: https://github.com/Pranav-player/Geometry-Visualizer-Web-